

Kaustubh Lall

Senior Machine Learning Engineer · GenAI & Agent Systems

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[GitHub](#) | [LinkedIn](#)

Software and machine learning engineer building production conversational agents, retrieval systems, and evaluation infrastructure. Healthcare is an application domain for my work across product engineering, cloud systems, and ML research.

EXPERIENCE

Senior Machine Learning Engineer | Feb 2024 - Present

ResMed

- Built Dawn largely from the ground up: a shared backend powering ResMed's generative-AI assistant in myAir and on the web, including both user-facing surfaces.
- Built Dusk, an AI-powered sleep coach with natural voice support, adaptive guidance, and personalized CBT-I plans.
- Built reusable RAG and agent infrastructure, Terraform-managed AWS environments, and LLM fine-tuning and evaluation pipelines; mentored engineers and reviewed code.

Founding Machine Learning & Cloud Engineer | 2021 - 2024

Radicle Science

- Built personalized participant reporting with Python and AWS Lambda, with SES/SNS for delivery and operational notifications.
- Developed NLP-assisted survey normalization, data pipelines, automated study metrics, and retention and recommendation systems for remote research studies.

Deep Learning Researcher | 2019 - 2021

UC San Diego · Radiology (Thornton Hospital)

- Built MRI segmentation and tissue-measurement workflows; packaged research pipelines with Docker and a Django scheduler for shared, reproducible use.
- Continued the collaboration through peer-reviewed shoulder and lumbar-spine imaging studies published in 2024 and 2025.

Software & ML Researcher | 2018 - 2020

UC San Diego · Leichtag Biomedical Center

- Built Python ML pipelines for drug-transporter research, combining interpretable baselines, exhaustive feature search, and NEAT-based feature engineering; coauthored a 2020 JBC paper.

SKILLS

Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, Segmentation (U-Net, SAM/MedSAM), LangChain & vector databases, Reproducible evaluation

Systems & Cloud: AWS (Lambda, SES/SNS, Elastic Beanstalk), Docker, Django, FastAPI, GraphQL

Languages: Python, TypeScript/JavaScript, Java, C/C++, SQL

Frontend: React, Next.js, Tailwind CSS

EDUCATION

UC San Diego | Graduate study, machine learning | 2020 - 2021

UC San Diego | Undergraduate study, engineering/computer science | 2016 - 2020

SELECTED PUBLICATIONS

Segment Anything Model (SAM) and Medical SAM (MedSAM) for Lumbar Spine MRI - Sensors, 2025.

Shoulder Bone Segmentation with DeepLab and U-Net - Osteology, 2024.

Unique metabolite preferences of the drug transporters OAT1 and OAT3 analyzed by machine learning - Journal of Biological Chemistry, 2020.